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UNIVERSITI PENDIDIKAN SULTAN IDRIS

SULTAN IDRIS EDUCATION UNIVERSITY

UNIVERSITI PENDIDIKAN SULTAN IDRIS

PENTAKSIRAN AKHIR
FINAL ASSESSMENT
SEMESTER 1 SESI 2020/2021
SEMESTER 1 SESSION 2020/2021

KOD / *CODE* : MES3103

KURSUS : PENGESAHAN DAN VERIFIKASI PERISIAN
COURSE :SOFTWARE VERIFICATION AND VALIDATION

ARAHAN / *INSTRUCTIONS*

1. Sila baca arahan pentaksiran dengan teliti.
   Please read the assessment instructions carefully.
2. Pelajar dikehendaki menyediakan skrip jawapan/tugasan/projek mengikut kepada pentaksiran akhir yang telah ditetapkan.
   Students are required to prepare the answer scripts/assignments/projects based on the predetermined the final assessment.
3. Pelajar dikehendaki menyerahkan skrip jawapan/tugasan/projek kepada pensyarah dalam tempoh masa yang telah ditetapkan.
   Students are required to submit their answer scripts/assignments/projects to the lecturers within the specified time.

PROGRAM / *PROGRAMME*: _____

TAHUN / *YEAR*: _____ KUMPULAN KULIAH / *CLASS GROUP*: _____

NO. PENDAFTARAN / *REGISTRATION NO*: _____

NO. KAD PENGENALAN / *I.C. NO*:

PENSYARAH / *LECTURER*: DR. BILAL BAHAA ZAIDAN

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SECTION A: (40 Marks)
Instruction: Answer **all** questions in a Microsoft word document, submission to be made through MyGuru.

1. Figure 1 shows an "Application registration" of a particular application that has three functions (i.e. sign in, signup and forget password). Function "Signup" required five arguments (email, username, password, name and birthday) while "sign in" function required two arguments (username and password). Finally, for "forget password" function, it requires email only. A software tester suggested you need to use "boundary value analysis" to test the program. Discuss the validity of his selection. Justify your answers.

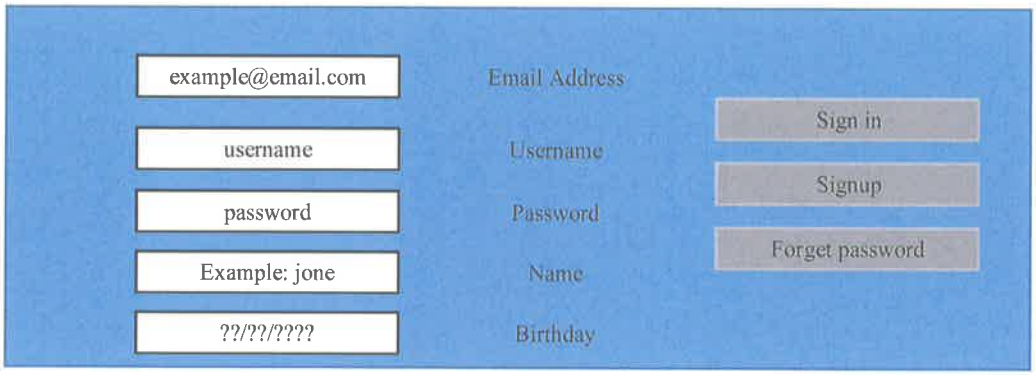

Figure 1: Application registration

[8 marks]

2. A health care center asks a software company to do smartphone apps assessment on 5 "patients follow-up" applications. Suggest the quality evaluation metrics that are required for the assessment associated with procedure of the assessment.

[8 marks]

3. The given code in Figure 2 is written to measure the factorial of an integer number using the recursion concept. Using the following code, answer the following question:

   a. What is the graph representation of the factorial function?

   [4marks]

   b. What are the main test cases required to test the factorial function based on black box testing technique?

   [4marks]

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```
#include<iostream>
using namespace std;

int factorial (int a)
{
        if(a<=1)
        {
                return 1;
        }
        else
                return (a*factorial(a-1));
}
void main()
{
        int x;
        cin>>x;
        cout<<factorial(x);
}
```

Figure 2

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4. The given programming code shown in figure 3 is written to perform the binary tree "insert" function. By using the following code, answer the following question:

   a) Transfer the graph below into data flow annotations.

   [8 marks]

   b) Is the graph represent the correct flow of the software? If no, please redraw the correct graph and justify your answers.

   [8 marks]

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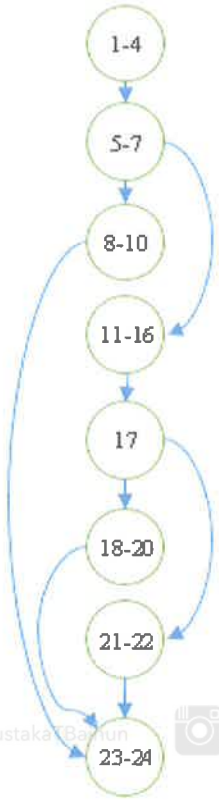
```
1   void btree::insert(tnode*&ptr,int item)
2       {
3       if (ptr==NULL)
4               {
5               ptr=new tnode;
6               if(ptr==NULL)
7                       {
8                       cout<<"can not allocate of memory"<<endl;
9                       }
10                      else
11                              {
12                              ptr->right=NULL;
13                              ptr->left=NULL;
14                              ptr->item=item;
15                              }
16              }
17              else
18                      {
19                      if (item<ptr->item)
20                      insert (ptr->left,item);
21                      else
22                      insert(ptr->right,item);
23                      }
24      }
```

The source code



Group

Figure 3

END OF QUESTIONS

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